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A Novel Data Fusion Method to Estimate Bridge Acceleration with Surrogate Inclination Mode Shapes through Independent Component Analysis

Xuzhao Lu [1,2], Chenxi Wei [1,2], Limin Sun [1,2,*], Ye Xia [1,2] and Wei Zhang [3,4]

[1] Department of Bridge Engineering, Tongji University, Shanghai 200092, China; luxuzhao_1992@tongji.edu.cn (X.L.); xichen@tongji.edu.cn (C.W.); yxia@tongji.edu.cn (Y.X.)
[2] Shanghai Qi Zhi Institute, Shanghai 200232, China
[3] Fujian Provincial Construction Engineering Quality Testing Center Co., Ltd., Fuzhou 350109, China; fjjkyzw@163.com
[4] Fujian Academy of Building Research Co., Ltd., Fuzhou 350109, China
* Correspondence: lmsun@tongji.edu.cn

Featured Application: This work can be applied to estimate acceleration using measured inclination, withoutpartially measuring the target acceleration. Moreover, this is a finiteelement model-free method and can be conveniently applied to beam bridges.

Abstract: Data fusion is an important issue in bridge health monitoring. Through data fusion, specific unknown bridge responses can be estimated with measured responses. However, existing data fusion methods always require a precise finite element model of the bridge or partially measured target responses, which are hard to realize in actual engineering. In this study, we propose a novel data fusion method. Measured inclinations across multiple cross-sections of the target bridge and accelerations at a subset of these sections were used to estimate accelerations at the remaining sections. Theoretical analysis of a typical vehicle-bridge interaction (VBI) system has shown parallels with the blind source separation (BSS) problem. Based on this, Independent Component Analysis (ICA) was applied to derive surrogate inclination mode shapes. This was followed by calculating surrogate displacement mode shapes through numerical integration. Finally, a surrogate inter-section transfer matrix for both measured and unmeasured accelerations was constructed, enabling the estimation of the target accelerations. This paper presents three key principles involving the relationship between the surrogate and actual inter-section transfer matrices, the integration of mode shape functions, and the consistency of transfer matrices for low- and high-frequency responses, which form the basis of the proposed method. A series of numerical simulations and a large-scale laboratory experiment were proposed to validate the proposed method. Compared to existing approaches, our proposed method stands out as a purely data-driven technique, eliminating the need for finite element analysis assessment. By incorporating the ICA algorithm and surrogate mode shapes, this study addresses the challenges associated with obtaining accurate mode shape functions from low-frequency responses. Moreover, our method does not require partial measurements of the target responses, simplifying the data collection process. The validation results demonstrate the method's practicality and convenience for real-world engineering applications, showcasing its potential for broad adoption in the field.

Keywords: bridge health monitoring; data fusion; independent component analysis (ICA); surrogate mode shape; vehicle–bridge interaction (VBI) system

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1. Introduction

In an optimal scenario for general bridge health monitoring systems, the ideal is for the bridge owner to have access to comprehensive response data (including displacement, acceleration, strain, inclination, etc.) across the entire bridge structure. This would enable a

more thorough assessment of the service status and precise detection of any damage [1,2]. However, budget constraints often limit the number of sensors that can be deployed to specific locations on the bridge. To address this challenge, scholars have recently turned to data fusion research [3], aiming to estimate unknown structural dynamics from the limited measured data available.

The key problem of data fusion is securing the transfer matrix that correlates mode shape functions with various response indicators, such as displacement [4], strain [5], inclination [6], and acceleration [7]. For example, one could compute the unknown acceleration at a specific site by simply multiplying the measured strain from different locations with a transfer matrix. However, the application of conventional mode shape identification techniques, such as the Stochastic Subspace Identification (SSI) [8] and eigensystem realization methods [9], is challenging for medium- and short-span bridges. Firstly, these methods struggle with identifying the low-frequency responses that predominantly affect strain and inclination measurements. Secondly, these methods necessitate a substantial number of sensors to maintain accuracy, which conflicts with the reality of limited sensor deployment. In light of these limitations, three main categories of data fusion methods have been proposed: those based on finite element models [10], Bayesian approaches [11], and machine learning techniques [12].

The finite element model-based approach necessitates an accurate finite element model of the target bridge [13,14]. This model serves as the foundation for calculating mode shape functions associated with various monitoring indices, including strain, inclination, and acceleration. It also facilitates the derivation of the transfer matrix that links these different indices. For example, Rapp et al. [15] developed a displacement–strain transfer matrix utilizing finite element models. An analogous strategy [16] involves estimating the external load from the finite element model and sensor responses, followed by a comprehensive finite element analysis to estimate the entire bridge's response. Subsequent research in this area has focused on two main aspects: (1) refining the model updating process [17] to enhance the accuracy of the finite element model and (2) advancing signal processing techniques [18] to mitigate measurement noise. A common de-noising technique involves integrating measured strain and acceleration data to estimate bridge displacement through methods such as regularization or Kalman filters [19]. Both strain and acceleration mode shapes are determined using finite element models. Moreover, the regularization process can be augmented with direct measurements obtained from advanced equipment like vision cameras [20] and millimeter wave radars [21], further enhancing the precision of the data fusion process.

However, constructing an accurate finite element model is challenging in practical engineering scenarios, as conventional model updating techniques can be exceedingly time-consuming. To address this issue, data fusion methods incorporating surrogate models have been introduced.

Bayesian-based methods offer an alternative to finite element modeling by constructing surrogate models, such as state-space models, directly from empirical data. For example [22], to infer bridge displacement, initial displacement estimates from a GPS device can be integrated with the double integration of measured accelerations to formulate a state-space model. These preliminary displacement measurements are instrumental in updating the state-space model. An alternative approach [23] forgoes direct measurements in favor of calculating the rotation of each structural segment from measured strain data, subsequently deriving displacement through cumulative summation. However, Bayesian methods invariably necessitate preliminary or approximate measurement of the target responses, i.e., before estimation, some true estimation results should be available, which can be a limitation given budget constraints. Thus, while Bayesian methods provide a valuable framework for data fusion, they still require a certain level of investment in advanced measurement technology or an adequate number of sensors, potentially conflicting with the budgetary restrictions often encountered in engineering projects.

An alternative approach involves the use of surrogate models through machine learning techniques [24–30]. These methods sidestep the construction of finite element or state-space models, opting instead to directly simulate a surrogate model. This model captures the transfer matrix that links monitoring data with the responses to be estimated. However, similar to Bayesian-based methods, machine learning approaches typically necessitate some level of measured responses at the target location. This requirement has become a focal point of research, often referred to as "missing data recovery" [31–33] or "response reconstruction" [34–36].

Despite these research efforts, current studies have yet to tackle scenarios where no sensors are installed at specific locations. Notably, some research [37] has suggested that machine learning can address this issue. It posits that a trained transfer matrix between strain and acceleration at one cross-section could be applied to other cross-sections, thus eliminating the need for partial measurements. However, this research has only been validated using a simply supported beam, where the strain and acceleration mode shapes are identical. This results in a consistent transfer matrix across all cross-sections, which is presumed to be a critical factor in validating the proposed method. For general beam bridges with multiple spans, the mode shapes for strain and acceleration may differ, leading to variations in the transfer matrix at different cross-sections. Consequently, the effectiveness of the method as demonstrated in this significant research may be questioned when applied to more complex bridge structures.

In this study, we proposed a novel finite element model-free method. Instead of a surrogate state-space model or directly trained surrogate model of transfer matrix with machine learning technologies, our method endeavors to derive surrogate mode shape functions. Through theoretical analysis of the dynamics of the simple vehicle–bridge interaction (VBI) system [38], we identified a parallel between the structural responses and the blind source separation (BSS) problem [39]. Leveraging this insight, we applied the Fast Independent Component Analysis (FastICA) algorithm [40], a prevalent technique for BSS, to compute surrogate inclination mode shapes. Further, this study employs discrete integration to deduce the surrogate displacement mode shape functions. These mode shape functions are then instrumental in formulating the surrogate inter-section transfer matrix for acceleration across various locations. A rigorous series of theoretical derivations elucidate three pivotal principles involving the identity between transfer matrices of low-frequency displacement and high-frequency acceleration, surrogate and real physical inter-section transfer matrices, and integration for the surrogate displacement mode shape function. These principles collectively validate the effectiveness of our proposed method. To confirm its reliability, we performed a series of numerical simulations and a full-scale laboratory test, demonstrating the method's practical applicability.

Compared to the SSI and eigenvalue methods, our proposed method overcomes the challenge of extracting modal information from low-frequency inclination data. As a finite element model-free approach, it offers high computational efficiency, making it more suitable for practical engineering applications. Additionally, unlike Bayesian-based and machine learning methods, it does not require partially measured responses at the target location, which is advantageous in budget-constrained scenarios. Moreover, in contrast to the aforementioned notable method, the effectiveness of our method has been confirmed for various types of beam bridges through a laboratory experiment involving a three-span continuous bridge.

The overall structure of the article is illustrated in Figure 1. In Section 2, a theoretical analysis of bridge dynamics in a simple VBI system was illustrated. Also, the whole framework of the proposed method with FastICA, discrete integration, and construction of inter-section transfer matrix is introduced. In Section 3, a validation process was conducted through a series of numerical analyses. In Section 4, the proposed method was validated through a large-scale laboratory experiment.

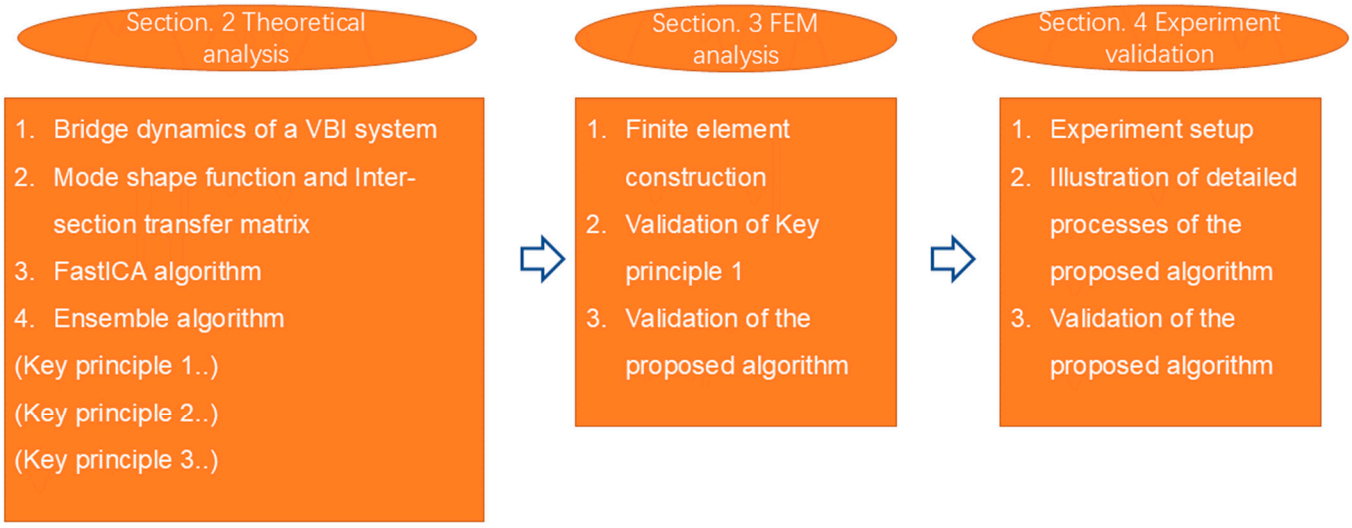


Figure 1. The overall structure of the whole study.

2. Theoretical Analysis

2.1. Bridge Dynamics of a Simple VBI System

A typical VBI system consists of a simply supported beam and a sprung-mass vehicle model (shown in Figure 2). The vehicle moves at a constant speed v.

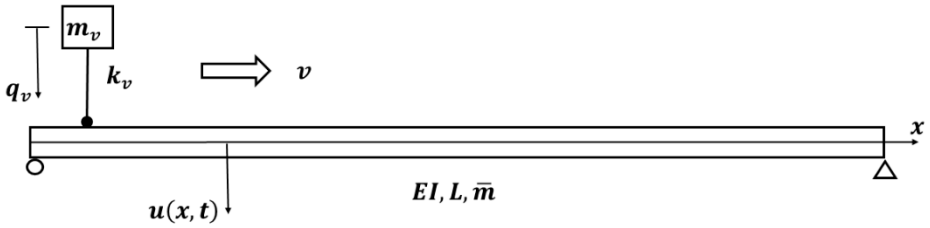


Figure 2. A simple VBI system [41].

The dynamic equation of the vehicle is [42]

$$m_v\ddot{q}_v + k_v q_v = k_v\, u|_{x=vt} \tag{1}$$

where m_v and k_v denote the sprung mass and the spring stiffness, respectively. q_v is the vertical displacement of the vehicle, and $\ddot{q}_v$ is the corresponding vehicle acceleration. $u|_{x=vt}$ is the vertical displacement of the contact point at x on the time point t.

The dynamic equation of the beam is [42]

$$\overline{m}\ddot{u} + EIu'''' = p(x,t) \tag{2a}$$

$$p(x,t) = [-m_v g + k_v\,(q_v - u\,|_{x=vt})]\delta(x - vt) \tag{2b}$$

where $\overline{m}$ denotes the unit mass of the bridge per meter. E and I are the stiffness and moment of inertia of the beam cross-section, respectively. L is the span of the beam. $p(x,t)$ is the interaction force applied on the bridge through the contact between the tire and bridge. $\ddot{u}$ is the second derivative of displacement with respect to time, and u'''' denotes the fourth derivative of displacement with respect to x. g is the gravitational acceleration, and $\delta(x - vt)$ is the Dirac delta function evaluated at the contact point $x = vt$ [42]. In the referenced study [43], it was determined that the amplitude of bridge vibrations caused by road roughness is significantly smaller compared to those caused by moving vehicles. Consequently, the impact of road roughness, being minimal, was considered negligible and thus excluded from our subsequent analysis. By assuming that the vehicle mass is much

smaller than the bridge mass, theoretical modal formulation for bridge dynamics can be derived as [42]

$$u(x,t) = \phi_{dis}(x)Y_{dis}(t) = \sum_n \frac{\Delta_{stn}}{1-S_n^2}\left\{ sin\frac{n\pi x}{L}\left[\sin\frac{n\pi vt}{L} - S_n \sin\omega_{bn}t\right]\right\} \quad (3)$$

where $\phi_{dis}(x)$ is the mode shape function, $Y_{dis}(t)$ is the time function, and ω_{bn} is the n-th bridge natural vibration frequency, and [42]

$$\Delta_{st,n} = \frac{-2m_v g L^3}{n^4\pi^4 EI} \quad (4a)$$

$$S_n = \frac{n\pi v}{L\omega_{b,n}} \quad (4b)$$

$$\omega_{b,n} = \frac{n^2\pi^2}{L^2}\sqrt{\frac{EI}{\overline{m}}} \quad (4c)$$

2.2. Mode Shape Function and Inter-Section Transfer Matrix of Bridge Displacement, Inclination, and Acceleration

The $u(x,t)$ in Equation (3) can be divided into a low-frequency component $u_{low}(x,t)$ (shown in Equation (5a)) and a high-frequency component $u_{high}(x,t)$ (shown in Equation (5b)) [42].

$$u_{low}(x,t) = \sum_n \frac{\Delta_{stn}}{1-S_n^2}\left[sin\frac{n\pi x}{L}\sin\frac{n\pi vt}{L}\right] \quad (5a)$$

$$u_{high}(x,t) = \sum_n \frac{\Delta_{stn}}{1-S_n^2}\left[-sin\frac{n\pi x}{L} S_n \sin\omega_{b,n}t\right] \quad (5b)$$

The division is based on the difference in a frequency range of these two components. The vibration frequencies of the low-frequency responses are $\frac{n\pi v}{L}$, which are the driving-force frequencies determined by the moving speed of the vehicle, while the frequencies of $u_{high}(x,t)$ are bridge natural vibration frequencies $\omega_{b,n}$. According to Yang and Lin [42] and Paultre [44], the frequencies of the first several driving-force modes are much lower than those of the first natural vibration modes. Moreover, displacement, inclination, and strain are mainly contributed by the first several low-frequency components, while acceleration is mainly contributed by the first several high-frequency components. Specifically, according to Equations (4a) and (5a), for the low-frequency responses, the ratio of contribution of each mode to the whole responses would be $1/n^4$ (i.e., $1/1$, $1/2^4$, $1/3^4 \ldots$) [41]. Therefore, to ensure the accuracy, the first two modes were considered in this study.

The displacement mode shape function is

$$\phi_{dis}(x) = sin\frac{n\pi x}{L} \quad (6)$$

which is the same for the low- and high-frequency responses. Assuming that three cross-sections were selected as x_1, x_2, and x_3 and only the first two driving-force modes are considered, one can obtain the following matrix (Equation (7)) from Equation (5a)):

$$\begin{bmatrix} sin\frac{\pi x_1}{L} & sin\frac{2\pi x_1}{L} \\ sin\frac{\pi x_2}{L} & sin\frac{2\pi x_2}{L} \\ sin\frac{\pi x_3}{L} & sin\frac{2\pi x_3}{L} \end{bmatrix} \begin{bmatrix} \frac{\Delta_{st,1}}{1-S_1^2} sin\frac{\pi v t_0}{L} & \frac{\Delta_{st,1}}{1-S_1^2} sin\frac{\pi v t_1}{L} & \cdots & \frac{\Delta_{st,1}}{1-S_1^2} sin\frac{\pi v t_N}{L} \\ \frac{\Delta_{st,2}}{1-S_2^2} sin\frac{2\pi v t_0}{L} & \frac{\Delta_{st,2}}{1-S_2^2} sin\frac{2\pi v t_1}{L} & \cdots & \frac{\Delta_{st,2}}{1-S_2^2} sin\frac{2\pi v t_N}{L} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\Delta_{st,n}}{1-S_n^2} sin\frac{n\pi v t_0}{L} & \frac{\Delta_{st,n}}{1-S_n^2} sin\frac{n\pi v t_1}{L} & \cdots & \frac{\Delta_{st,n}}{1-S_n^2} sin\frac{n\pi v t_N}{L} \end{bmatrix} = \boldsymbol{\Phi}(\boldsymbol{x}) \cdot \boldsymbol{Y}(\boldsymbol{t}) = \begin{bmatrix} u_{x_1,t_0} & u_{x_1,t_1} & \cdots & u_{x_1,t_N} \\ u_{x_2,t_0} & u_{x_2,t_1} & \cdots & u_{x_2,t_N} \\ u_{x_3,t_0} & u_{x_3,t_1} & \cdots & u_{x_3,t_N} \end{bmatrix} \tag{7}$$

where $t_0, t_1, \ldots t_N$ are the $N + 1$ time points. Further, two sub-equations (Equation (8a,b)) can be generated from Equation (7).

$$\begin{bmatrix} sin\frac{\pi x_1}{L} & sin\frac{2\pi x_1}{L} \\ sin\frac{\pi x_2}{L} & sin\frac{2\pi x_2}{L} \end{bmatrix} \begin{bmatrix} \frac{\Delta_{st,1}}{1-S_1^2} sin\frac{\pi v t_0}{L} & \frac{\Delta_{st,1}}{1-S_1^2} sin\frac{\pi v t_1}{L} & \cdots & \frac{\Delta_{st,1}}{1-S_1^2} sin\frac{\pi v t_N}{L} \\ \frac{\Delta_{st,2}}{1-S_2^2} sin\frac{2\pi v t_0}{L} & \frac{\Delta_{st,2}}{1-S_2^2} sin\frac{2\pi v t_1}{L} & \cdots & \frac{\Delta_{st,2}}{1-S_2^2} sin\frac{2\pi v t_N}{L} \end{bmatrix} = \begin{bmatrix} u_{x_1,t_0} & u_{x_1,t_1} & \cdots & u_{x_1,t_N} \\ u_{x_2,t_0} & u_{x_2,t_1} & \cdots & u_{x_2,t_N} \end{bmatrix} \tag{8a}$$

$$\begin{bmatrix} sin\frac{\pi x_2}{L} & sin\frac{2\pi x_2}{L} \\ sin\frac{\pi x_3}{L} & sin\frac{2\pi x_3}{L} \end{bmatrix} \begin{bmatrix} \frac{\Delta_{st,1}}{1-S_1^2} sin\frac{\pi v t_0}{L} & \frac{\Delta_{st,1}}{1-S_1^2} sin\frac{\pi v t_1}{L} & \cdots & \frac{\Delta_{st,1}}{1-S_1^2} sin\frac{\pi v t_N}{L} \\ \frac{\Delta_{st,2}}{1-S_2^2} sin\frac{2\pi v t_0}{L} & \frac{\Delta_{st,2}}{1-S_2^2} sin\frac{2\pi v t_1}{L} & \cdots & \frac{\Delta_{st,2}}{1-S_2^2} sin\frac{2\pi v t_N}{L} \end{bmatrix} = \begin{bmatrix} u_{x_2,t_0} & u_{x_2,t_1} & \cdots & u_{x_2,t_N} \\ u_{x_3,t_0} & u_{x_3,t_1} & \cdots & u_{x_3,t_N} \end{bmatrix} \tag{8b}$$

The responses at x_3 can then be derived from Equation (8a,b) as

$$\begin{bmatrix} u_{x_2,t_0} & u_{x_2,t_1} & \cdots & u_{x_2,t_N} \\ u_{x_3,t_0} & u_{x_3,t_1} & \cdots & u_{x_3,t_N} \end{bmatrix} = T_{dis} \begin{bmatrix} u_{x_1,t_0} & u_{x_1,t_1} & \cdots & u_{x_1,t_N} \\ u_{x_2,t_0} & u_{x_2,t_1} & \cdots & u_{x_2,t_N} \end{bmatrix} \tag{9}$$

where T_{dis} is the inter-section transfer matrix of low-frequency displacement from x_1 and x_2 to x_3, and

$$T_{dis} = \begin{bmatrix} sin\frac{\pi x_2}{L} & sin\frac{2\pi x_2}{L} \\ sin\frac{\pi x_3}{L} & sin\frac{2\pi x_3}{L} \end{bmatrix} \begin{bmatrix} sin\frac{\pi x_1}{L} & sin\frac{2\pi x_1}{L} \\ sin\frac{\pi x_2}{L} & sin\frac{2\pi x_2}{L} \end{bmatrix}^{-1} \tag{10}$$

Considering that the inclination and acceleration can be calculated following Equation (11a,b), we obtain

$$\theta_{low}(x,t) = \frac{du_{low}(x,t)}{dx} = \frac{d\phi_{dis}(x)}{dx} \cdot Y_{dis-low}(t) \tag{11a}$$

$$\ddot{u}_{high}(x,t) = \frac{d^2 u_{high}(x,t)}{dt^2} = \phi_{dis}(x) \cdot \frac{d^2 Y_{dis-high}(t)}{dt^2} \tag{11b}$$

The inclination and acceleration mode shape functions can be calculated by substituting Equation (5a) on to Equation (11a,b), respectively, as

$$\phi_{incl}(x) = cos\frac{n\pi x}{L} \tag{12a}$$

and

$$\phi_{acc}(x) = sin\frac{n\pi x}{L} \tag{12b}$$

There are two features that should be noted:

(1) Equation (11a) demonstrates that the inclination mode shape function is the derivative of the displacement mode shape function with respect to x, which should also hold true for continuous beams. Moreover, the time function $Y_{dis-low}(t)$ is identical for inclination and displacement.

(2) Equation (11b) illustrates that the process of differentiating displacement twice to obtain acceleration does not alter the mode shape function.

In detail, the inter-section transfer matrices for the inclination and acceleration are

$$T_{incl} = \begin{bmatrix} cos\frac{\pi x_2}{L} & cos\frac{2\pi x_2}{L} \\ cos\frac{\pi x_3}{L} & cos\frac{2\pi x_3}{L} \end{bmatrix} \begin{bmatrix} cos\frac{\pi x_1}{L} & cos\frac{2\pi x_1}{L} \\ cos\frac{\pi x_2}{L} & cos\frac{2\pi x_2}{L} \end{bmatrix}^{-1} \tag{13a}$$

and

$$T_{acc} = T_{dis} = \begin{bmatrix} sin\frac{\pi x_2}{L} & sin\frac{2\pi x_2}{L} \\ sin\frac{\pi x_3}{L} & sin\frac{2\pi x_3}{L} \end{bmatrix} \begin{bmatrix} sin\frac{\pi x_1}{L} & sin\frac{2\pi x_1}{L} \\ sin\frac{\pi x_2}{L} & sin\frac{2\pi x_2}{L} \end{bmatrix}^{-1} \tag{13b}$$

Then, ***the first key principle*** can be derived from Equations (12b) and (13b):

First key principle:*The high-frequency acceleration mode shape function and low-frequency displacement mode shape function are identical. Consequently, transfer matrices for the low-frequency displacement and high-frequency acceleration are identical.*

2.3. FastICA for Surrogate Inclination Mode Shape Function

The FastICA algorithm is a prevalent technique for addressing the BSS problem, which involves estimating the unknown mixing matrix $\mathbf{A}$ and m independent signal sources s_m that combine to produce the measured signal matrix $\mathbf{z}$, as depicted in Equation (14a,b) [45]:

$$\mathbf{z} = \mathbf{As} \tag{14a}$$

in which

$$\mathbf{s} = (s_1, s_2, \ldots, s_m)^{\boldsymbol{T}} \tag{14b}$$

The expansion of Equation (14a) is

$$\begin{bmatrix} z_{1,t_0} & z_{1,t_1} & \cdots & z_{1,t_N} \\ z_{2,t_0} & z_{2,t_1} & \cdots & z_{2,t_N} \\ \vdots & \vdots & \vdots & \vdots \\ z_{O,t_0} & z_{O,t_1} & \cdots & z_{O,t_N} \end{bmatrix} = \begin{bmatrix} A_{1,1} & A_{1,2} & \cdots & A_{1,m} \\ A_{2,1} & A_{2,2} & \cdots & A_{2,m} \\ \vdots & \vdots & \vdots & \vdots \\ A_{O,1} & A_{O,2} & \cdots & A_{O,m} \end{bmatrix} \begin{bmatrix} s_{1,t_0} & s_{1,t_1} & \cdots & s_{1,t_N} \\ s_{2,t_0} & s_{2,t_1} & \cdots & s_{2,t_N} \\ \vdots & \vdots & \vdots & \vdots \\ s_{m,t_0} & s_{m,t_1} & \cdots & s_{m,t_N} \end{bmatrix} \tag{15}$$

in which the subscript 1, 2, ... O denote O-th signal receivers.

The resemblance between structural dynamics and the BSS problem can be discerned by examining the parallels between Equations (3) and (14a), as well as Equations (6) and (15). In both cases, the measured signals $\mathbf{z}$ and the displacement (also for inclination and acceleration) at multiple cross-sections are the result of a weighting matrix ($\mathbf{A}$ and $\boldsymbol{\Phi}(\boldsymbol{x})$) multiplying a source signal matrix ($\mathbf{s}$ and $\boldsymbol{Y}(\boldsymbol{t})$). Consequently, in this study, we have harnessed the FastICA algorithm [40] to derive a surrogate inclination mode shape function from the dynamic inclinations measured at multiple cross-sections.

However, the FastICA ensures only the statistical independence of the separated signal sources in $\mathbf{s}$. The separated matrices $\mathbf{A}$ and $\mathbf{s}$ may not correspond directly to the physical model of inclination shapes and their associated time functions. For instance, if the measured inclination at x_1 is mainly contributed by the first two vibration modes, then the inclination at x_1 can be formulated as

$$\mathbf{z_{incl}}(x_1, t) = \phi_{incl,1}(x_1) \cdot Y_{incl,1}(t) + \phi_{incl,2}(x_1) \cdot Y_{incl,2}(t) \tag{16}$$

where $\phi_{incl,1}(x_1)$ and $\phi_{incl,2}(x_1)$ denote the value of the inclination mode shape functions $\phi_{incl,1}(x)$ and $\phi_{incl,2}(x)$ at x_1, while the FastICA separation result might be

$$\mathbf{z_{incl}}(x_1, t) = \phi'_1(x_1) \cdot s_1(t) + \phi'_2(x_1) \cdot s_2(t) \tag{17}$$

for which

$$s_1(t) = a \cdot Y_{incl,1}(t) + b \cdot Y_{incl,2}(t) \tag{18a}$$

$$s_2(t) = c \cdot Y_{incl,1}(t) + d \cdot Y_{incl,2}(t) \tag{18b}$$

where a, b, c, and d, are unknown weights. $\phi'_1(x_1)$ and $\phi'_2(x_1)$ are vectors in **A**. These two vectors are used as surrogate mode shape functions. Also, $s_1(t)$ and $s_2(t)$ are separated independent signal series. Furthermore, a relationship can be established between the actual and surrogate mode shape coordinates as follows:

$$\left(\phi'_1(x_1), \phi'_2(x_1)\right) T_{S-R} = \left(\phi_{incl,1}(x_1), \phi_{incl,2}(x_1)\right) \tag{19a}$$

where T_{S-R} is the transfer matrix between the surrogate mode shape vector to the real mode shape vector,

$$T_{S-R} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \tag{19b}$$

Moreover, $s_1(t)$ and $s_2(t)$ are consistent across all the cross-sections, ensuring that Equations (18a) and (18b) hold true for each cross-section. Consequently, Equation (19a) can be generalized to represent the entire bridge as follows:

$$\left(\phi'_1(x), \phi'_2(x)\right) T_{S-R} = \left(\phi_{incl,1}(x), \phi_{incl,2}(x)\right) \tag{20a}$$

Additionally, the surrogate inter-section transfer matrix for bridge inclination can be derived as Equation (20b),

$$T_{sur-incl} = \begin{bmatrix} \phi_{incl,\,2,1} & \phi_{incl,\,2,2} \\ \phi_{incl,\,3,1} & \phi_{incl,\,3,2} \end{bmatrix} {T_{S-R}}^{-1} T_{S-R} \begin{bmatrix} \phi_{incl,\,1,1} & \phi_{incl,\,1,2} \\ \phi_{incl,\,2,1} & \phi_{incl,\,2,2} \end{bmatrix}^{-1} = \begin{bmatrix} \phi_{incl,\,2,1} & \phi_{incl,\,2,2} \\ \phi_{incl,\,3,1} & \phi_{incl,\,3,2} \end{bmatrix} \begin{bmatrix} \phi_{incl,\,1,1} & \phi_{incl,\,1,2} \\ \phi_{incl,\,2,1} & \phi_{incl,\,2,2} \end{bmatrix}^{-1} = T_{incl} \tag{20b}$$

from which ***the second key principle*** has been generated:

Second key principle:*The surrogate inter-section transfer matrix obtained from FastICA is exactly identical to the real physical transfer matrix of inclination.*

Additionally, according to Equation (20b), there is no need to calculate the T_{S-R} matrix and its elements. Furthermore, the second key principle should also apply to other response indices, such as displacement and acceleration.

2.4. Algorithm

In this research, our objective is to estimate the high-frequency acceleration at a specific location on the target bridge, where only an inclinometer is installed. Inclinometers and accelerometers are, however, placed at several other locations. The collected inclination and acceleration data from these locations will be utilized to deduce the unknown acceleration at the target site. The flowchart illustrating the complete algorithm is presented in Figure 3.

$(\theta_{x_1}(t), \theta_{x_2}(t) \ldots \theta_{x_p}(t)$ denotes the measured inclination time histories at p-th cross-sections x_1, $x_2, \ldots x_p$; as an example, it is assumed that at x_1 and x_2, the acceleration has been measured (as $\ddot{u}_{x_1}(t)$ and $\ddot{u}_{x_2}(t)$, respectively), and the accelerations at these two cross-sections are used to estimate the unmeasured acceleration at x_3 (i.e., $\ddot{u}_{x_3}(t)$).

Step 1: Inclination data from multiple cross-sections are subjected to analysis using the FastICA algorithm to extract the mixing matrix W_{mix} and the discrete surrogate inclination mode shape functions ϕ'_{incl-1} and ϕ'_{incl-2}.

Focusing on the first two modes, we assume the presence of only two independent signal sources, s_1 and s_2. Should additional modes be taken into account, the number of signal sources would correspondingly expand to match the total count of modes. For each cross-section acting as a signal receiver, there would be two corresponding separated weight parameters, w_1 and w_2. Assuming that there are p cross-sections, the mixing matrix would be

$$W_{mix} = \begin{bmatrix} w_{1,1} & w_{2,1} & \ldots & w_{p,1} \\ w_{1,2} & w_{2,2} & \ldots & w_{p,2} \end{bmatrix} \tag{21a}$$

and the discrete first surrogate inclination mode shape function would be the first line of W_{mix},

$$\phi'_{incl-1} = \begin{bmatrix} w_{1,1} & w_{2,1} & \ldots & w_{p,1} \end{bmatrix} \tag{21b}$$

and similarly, the second surrogate inclination mode shape function would be

$$\phi'_{incl-2} = \begin{bmatrix} w_{1,2} & w_{2,2} & \ldots & w_{p,2} \end{bmatrix} \tag{21c}$$

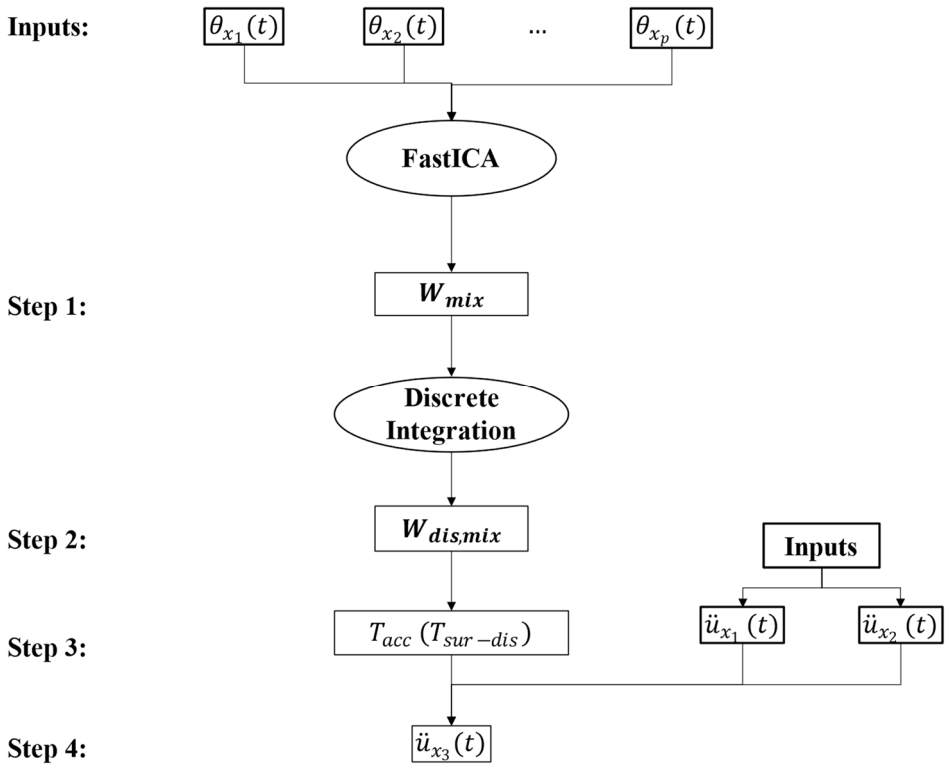


Figure 3. Flowchart of the algorithm.

Step 2: Discrete integration of surrogate inclination mode shapes to generate surrogate displacement mode shapes.

The nearest two supports are set as location x_0 and location x_{p+1}. Coordinates of these $p + 2$ locations are

$$x = \begin{bmatrix} x_0 & x_1 & \dots & x_{p+1} \end{bmatrix} \tag{22}$$

Considering that the displacement at the bridge supports is inherently zero, the displacement mode shape coordinates at these supports are accordingly set to zero. To generate the surrogate displacement mode shape function for these intermediate p coordinates, we employ discrete integration techniques: the Euler method for the endpoints x_1 and x_p and Hermite polynomials for the internal range from x_2 to x_{p-1}. This approach yields the surrogate displacement mode shape functions at the intermediate p coordinates as follows:

$$\phi'_{dis-1} = \begin{bmatrix} w_{dis,\,1,1} & w_{dis,\,2,1} & \dots & w_{dis,\,p,1} \end{bmatrix} \tag{23a}$$

$$\phi'_{dis-2} = \begin{bmatrix} w_{dis,\,1,2} & w_{dis,\,2,2} & \dots & w_{dis,\,p,2} \end{bmatrix} \tag{23b}$$

and ϕ'_{dis-1} and ϕ'_{dis-2} form the new mixing matrix for bridge displacement as

$$W_{dis,mix} = \left[\phi'_{dis-1}, \phi'_{dis-2}\right] \tag{23c}$$

Additionally, as indicated by Equation (20a), the transfer matrix T_{S-R} in Equation (20a) is irrelevant to the spatial coordinate x; then, Equation (24a) can be generated as

$$W_{dis,mix} T_{S-R} = \boldsymbol{\Phi_{dis}}(\boldsymbol{x}) \tag{24a}$$

in which

$$\boldsymbol{\Phi_{dis}}(\boldsymbol{x}) = \left[\phi_{dis,1}(x), \phi_{dis,2}(x)\right] \tag{24b}$$

where $\phi_{dis,1}(x)$ and $\phi_{dis,2}(x)$ are the real displacement mode shape functions.

Step 3: Construct an inter-section transfer matrix for bridge displacement.

Assume that both acceleration and inclination at cross-sections x_1 and x_2 were measured, while at x_3, only the inclination was measured. According to Equations (23a) and (23b), a surrogate inter-

section transfer matrix for displacement would be constructed with the surrogate displacement mode shape functions following Equation (10) as

$$T_{sur-dis} = \begin{bmatrix} w_{dis,\,2,1} & w_{dis,\,2,2} \\ w_{dis,\,3,1} & w_{dis,\,3,2} \end{bmatrix} \begin{bmatrix} w_{dis,\,1,1} & w_{dis,\,1,2} \\ w_{dis,\,2,1} & w_{dis,\,2,2} \end{bmatrix}^{-1} \tag{25}$$

Substituting Equation (24a) onto Equation (25), one can obtain

$$T_{sur-dis} = \begin{bmatrix} \phi_{dis,\,2,1} & \phi_{dis,\,2,2} \\ \phi_{dis,\,3,1} & \phi_{dis,\,3,2} \end{bmatrix} {T_{S-R}}^{-1} T_{S-R} \begin{bmatrix} \phi_{dis,\,1,1} & \phi_{dis,\,1,2} \\ \phi_{dis,\,2,1} & \phi_{dis,\,2,2} \end{bmatrix}^{-1} = \begin{bmatrix} \phi_{dis,\,2,1} & \phi_{dis,\,2,2} \\ \phi_{dis,\,3,1} & \phi_{dis,\,3,2} \end{bmatrix} \begin{bmatrix} \phi_{dis,\,1,1} & \phi_{dis,\,1,2} \\ \phi_{dis,\,2,1} & \phi_{dis,\,2,2} \end{bmatrix}^{-1} = T_{dis} \tag{26}$$

in which T_{dis} is the real displacement inter-section transfer matrix. Equation (26) reveals the ***third key principle***:

Third key principle:*The surrogate inter-section transfer matrix for displacement, derived from the integration of the surrogate inclination mode shape functions, is equivalent to the actual inter-section transfer matrix for displacement.*

Step 4: Since the first key principle (illustrated as Equation (13b)) has shown that $T_{dis} = T_{acc}$, it can be inferred from Equation (26) that

$$T_{sur-dis} = T_{acc} \tag{27}$$

Then, one can estimate the acceleration at x_3 using the measured acceleration at x_1 and x_2:

$$\begin{bmatrix} \ddot{u}_{x_2,t_0} & \ddot{u}_{x_2,t_1} & \cdots & \ddot{u}_{x_2,t_N} \\ \ddot{u}_{x_3,t_0} & \ddot{u}_{x_3,t_1} & \cdots & \ddot{u}_{x_3,t_N} \end{bmatrix} = T_{sur-dis} \begin{bmatrix} \ddot{u}_{x_1,t_0} & \ddot{u}_{x_1,t_1} & \cdots & \ddot{u}_{x_1,t_N} \\ \ddot{u}_{x_2,t_0} & \ddot{u}_{x_2,t_1} & \cdots & \ddot{u}_{x_2,t_N} \end{bmatrix} \tag{28}$$

Additionally, it is important to note that throughout the entire algorithm, the T_{S-R} matrix remains unknown and has effectively been eliminated from the process. Consequently, there is no need to determine the explicit form of T_{S-R}.

It should be noted that once the T_{acc} has been determined for a specific scenario under a given traffic loading condition, it should also be applicable to other scenarios with different traffic loadings. This is because, as indicated in Equation (26), the T_{acc} is solely dependent on the bridge's mode shapes, which are independent of different traffic loadings. This assertion has been substantiated by previous research, and further details can be found in reference [41]. Should a bridge experience degradation that leads to a significant and uneven reduction in its stiffness along its span [46], this would inevitably alter the bridge's mode shapes. Under such circumstances, it is presumed that the T_{acc} (transfer matrix for acceleration) derived from a pristine bridge would no longer be suitable for the compromised structure. Nevertheless, if monitoring data are gathered from the degraded bridge, our proposed method can be effectively employed to ascertain a new T_{acc} that aligns with the modified mode shapes of the deteriorated bridge.

3. Finite Element Simulation

3.1. Finite Element Model

A finite element model of a simple VBI system has been constructed to validate the proposed method. The VBI system adheres to the configuration established in a prior study [42]. The bridge is a simply supported beam with a span of 25 m, and its physical parameters are illustrated in Table 1. A simplified quarter-car model is employed to simulate the vehicle, encompassing a sprung mass, a suspension spring, and a wheel interface in contact with the beam's surface. The comprehensive parameters of the vehicle are enumerated in Table 2.

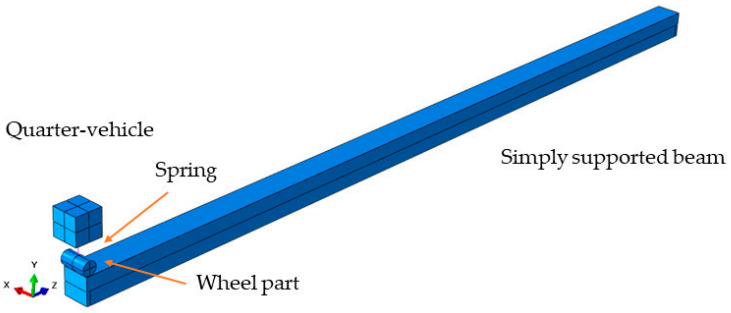
Table 1. Physical parameters of the simply supported beam [42].

Sprung mass (kg)	600
Spring stiffness (N/m)	500,000
Bouncing frequency (Hz)	4.60

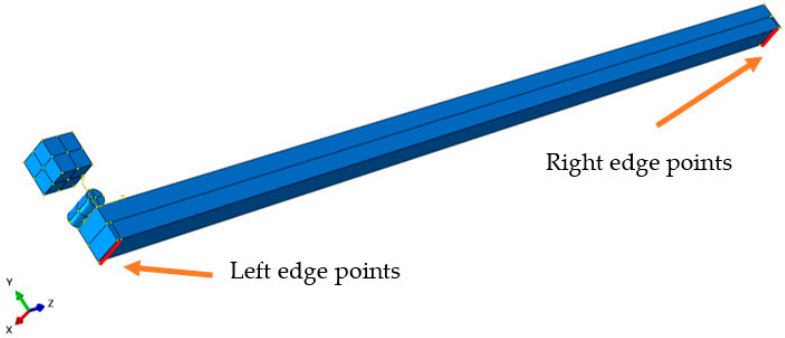
Table 2. Parameters of the vehicle model [42].

Mass per meter (kg/m)	4800
EI (N/m^2)	3.33×10^9
First bending frequency (Hz)	2.08
Second bending frequency (Hz)	8.33
Length (m)	25

A finite element model, depicted in Figure 4a, was constructed utilizing the commercial software ABAQUS 2020 [47]. Solid elements of type C3D8R were utilized to model both the beam and the sprung mass, with each element measuring 0.1 m × 0.1 m × 0.1 m. Our preliminary analysis confirmed that the mesh density exerts a negligible impact on the simulation's accuracy. The boundary conditions (Figure 4b) for the simply supported beam were implemented by restraining the appropriate degrees of freedom at the beam's edge nodes. For the left edge nodes, which are part of a simply supported beam, movements along the X, Y, and Z axes, as well as rotations about the Y and Z axes, were constrained. Similarly, for the right edge nodes, constraints were applied to motions along the X and Y axes and rotations about the Y and Z axes. The spring element was used to model the connection between the sprung mass and the wheel. The wheel, depicted as a cylindrical solid element, engaged with the beam surface using the penalty method. It is important to note that during the simulation, the wheel's rotation was restricted. The friction coefficient between the wheel and the bridge surface was set to zero to simplify the interaction. Consistent with the theoretical derivation and considering its minimal impact on bridge dynamics, road roughness was excluded from the analysis. The vehicle's movement across the bridge was assumed to be at uniform speeds. The simulation was divided into two computational stages: a static analysis to account for gravitational forces on the entire VBI system, followed by a dynamic analysis to assess the system's responses to the moving vehicle. The dynamic analysis was conducted using the Hilber–Hughes–Taylor (HHT) algorithm with a time step of 0.01 s. The simulation approach we utilized is grounded in the method initially proposed and substantiated in the existing body of research [48]. For an exhaustive examination of the simulation methodology, one can refer to the detailed account provided in the aforementioned study [48].



(**a**) Upper view



(**b**) Bottom view (Edge points are constrained to simulate boundary conditions and marked with red lines.)

Figure 4. Finite element model of the VBI system [41].

3.2. Result Analysis

In this study, according to the theoretical derivation from Equations (4a) and (5a), it is hypothesized that the first two vibration modes dominate the bridge responses. Our approach involved estimating the acceleration at a particular point on the beam using acceleration data from two distinct locations. Adhering to the second key principle, we utilized the displacement histories captured at various points to generate two surrogate displacement mode shapes via the FastICA algorithm. Subsequently, in accordance with the first key principle as outlined in Step 4, these surrogate shapes were employed to deduce the target acceleration.

For the initial scenario, the vehicle's speed was set at 5 m/s. A selection of seven observation points was made, with their respective coordinates detailed in Table 3. A representative displacement time history is illustrated in Figure 5. The FastICA algorithm was tailored to identify two independent signal sources, corresponding to the first two surrogate vibration modes of the bridge, with signals captured at seven designated receivers, each corresponding to the specific locations where displacement measurements were recorded.

Table 3. Coordinates of seven selected observation points.

Points	1	2	3	4	5	6	7
Coordinate (m)	1.2	6.9	9.7	13	16	19	23.6

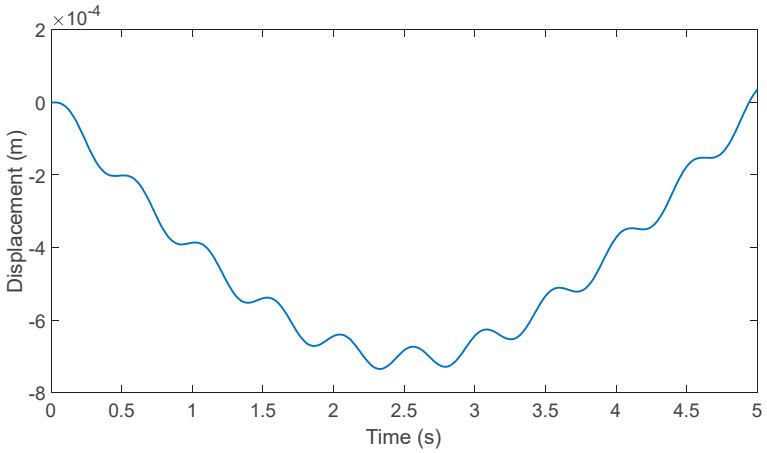


Figure 5. Displacement time history at Point 3.

All seven displacement time histories were utilized as inputs for the FastICA algorithm. From the mixing matrix, the FastICA yielded two distinct surrogate displacement mode shape curves, which are presented in Figure 6. It is evident that these two curves deviate from the actual first and second mode shape functions of the bridge, which are sinusoidal functions represented by $sinx$ and $sin2x$, respectively.

Subsequently, the proposed method's effectiveness was evaluated. The accelerations recorded at Points 7 and 4 were utilized to compute the surrogate inter-section transfer matrix of acceleration, which was formulated based on the two surrogate displacement mode shapes. Utilizing this matrix, the acceleration at Point 6 was deduced. The estimated acceleration at Point 6 was then compared with the results from the simulation, as displayed in Figure 7a for the time domain analysis, and in Figure 7b for the frequency domain analysis.

In the time domain, as shown in Figure 7a, the estimated acceleration time history aligns closely with the simulation outcome. However, between 4.5 s and 5 s, when the vehicle is positioned above Point 7, it generates high-frequency numerical noise. This noise is transmitted to the estimation result via the proposed method, leading to a noticeable deviation between the estimated and simulated time histories at Point 6 during this interval.

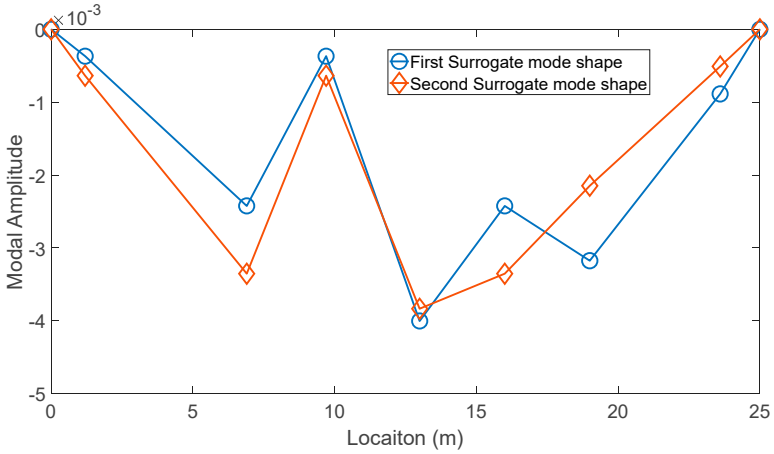
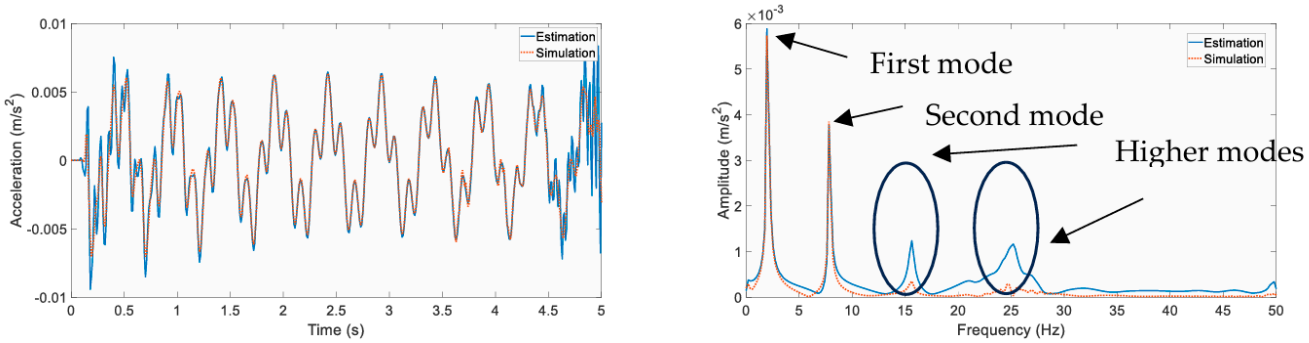


Figure 6. Two surrogate displacement mode shapes from the FastICA mixing matrix. (The first surrogate mode shape is indicated with circles, while the second is denoted by diamonds, across the seven monitored sites and at both end supports).

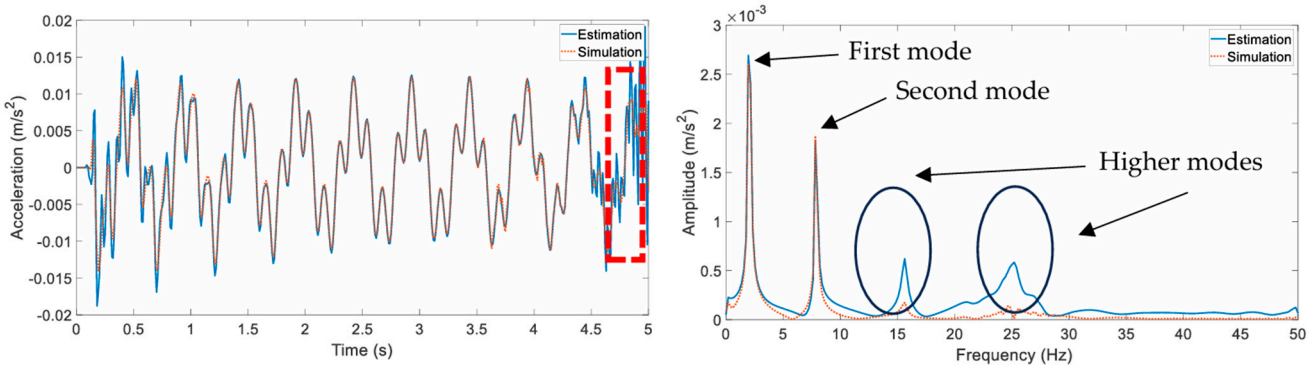


(**a**) Time domain (**b**) Frequency domain

Figure 7. Comparison between the estimated acceleration through the proposed method and the simulation result at Point 6 (higher modes are marked with ellipses and similarly marked below).

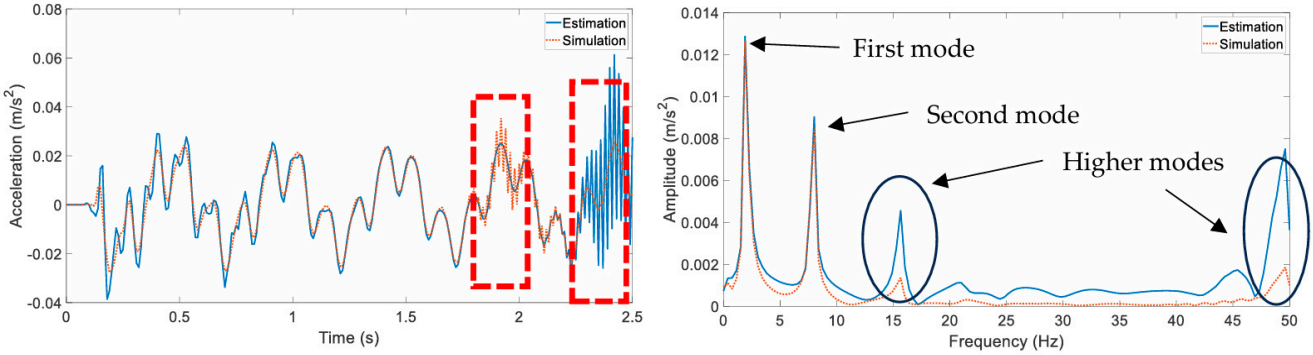
Furthermore, in the frequency domain comparison depicted in Figure 7b, the estimated and simulation results are in good agreement for the first and second vibration modes, thereby confirming the validity of the proposed algorithm. Nonetheless, the high-frequency components—highlighted with ellipses in Figure 7b—exhibit significant discrepancies. This discrepancy arises because the current study focuses solely on the first two vibration modes based on acceleration data from two specific locations, limiting the effective consideration of higher-order modes.

The inter-section transfer matrix for bridge acceleration, derived from the proposed method, was further validated across various scenarios. As established in the previous study [41], the transfer matrix would remain invariant under diverse moving conditions. To validate the proposed method, two additional scenarios were modeled, one with a vehicle mass doubled and the other with the vehicle's speed doubled to 10 m/s. It should be noted that, for the speeds of 5 m/s and 10 m/s, they correspond to 18 km/h and 36 km/h, which are common vehicle running speeds in urban transportation systems. The transfer matrix also kept constant with even higher moving speeds. The acceleration at Point 6 in these new scenarios was estimated by applying the original scenario's transfer matrix to the simulated accelerations of these scenarios. Figures 8 and 9 present the comparative analysis between the estimated and simulated results. The consistency observed in both the time and frequency domains substantiates that the obtained inter-section transfer matrix is equivalent to the actual transfer matrix, thereby validating the proposed method.



(**a**) Time domain (rectangle marks the inconsistencies) (**b**) Frequency domain

Figure 8. Comparison between the estimated acceleration using the proposed method and the simulation result at Point 6 in Scenario 2 with double vehicle mass.



(**a**) Time domain (rectangles mark the inconsistencies) (**b**) Frequency domain

Figure 9. Comparison between the estimated acceleration using the proposed method and the simulation result at Point 6 in Scenario 2 with higher moving speed.

In addition, discrepancies are evident in Figures 8 and 9, manifesting in both the time and frequency domains. As depicted in Figure 8b, these inconsistencies are particularly pronounced in the vibrations of higher modes. It is assumed that in this study, we have focused exclusively on the first two vibration modes, and accordingly, the number of signal sources has been set to two. Consequently, the proposed algorithm automatically isolates two independent signal sources along with their corresponding mixing matrix. As established in Section 2.2, the first two modes predominantly influence the overall bridge response; thus, the two signal sources and the mixing matrix should be primarily associated with these initial two modes. However, for vibrations of higher modes, which are distinct from the first two modes, the current algorithm's configuration, which only accounts for two signal sources, is insufficient for accurate estimation. To achieve precise estimation of higher modes, which necessitates the consideration of more independent signal sources, the algorithm's parameter settings, specifically the number of signal sources, should be reevaluated. Furthermore, acceleration measurements should be collected from a greater number of cross-sections, rather than the current limitation of just two, which corresponds to the available signal resources.

4. Experiment Validation

4.1. Experiment Set-Up

In this study, to validate the proposed method, a moving-vehicle laboratory experiment was executed on a large-scale steel continuous beam bridge. The bridge features a span arrangement totaling 22 m, composed of segments measuring 6, 10, and 6 m, respectively, as depicted in Figure 10. The southwest end support is equipped with a fixed bearing, whereas the remaining supports are sliding bearings. Steel platforms, illustrated in Figure 11, have been installed at both ends of the bridge to serve as starting and stopping points for the vehicle. The bridge's cross-sectional details are presented in Figure 12. The bridge's fundamental and second bending frequencies are recorded at 9.854 Hz and 20.015 Hz, respectively.

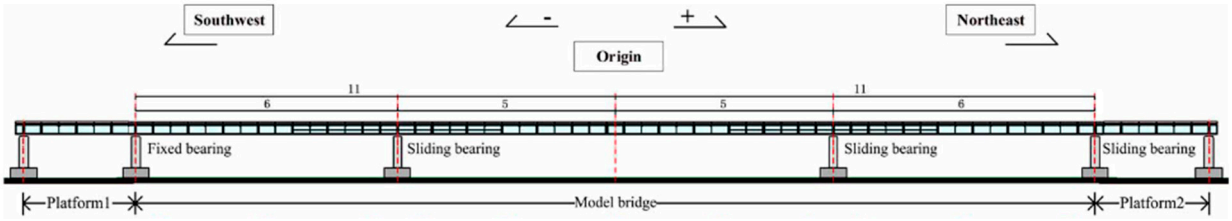


Figure 10. Three-span steel continuous bridge [16].

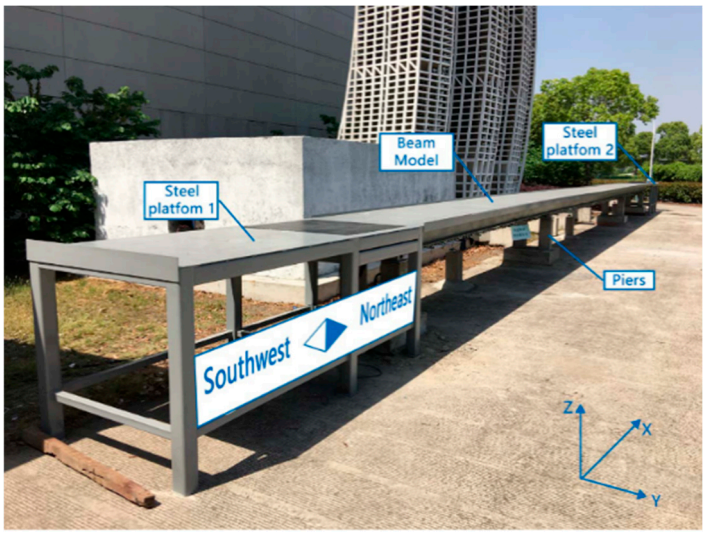


Figure 11. Steel platform and steel bridge [16].

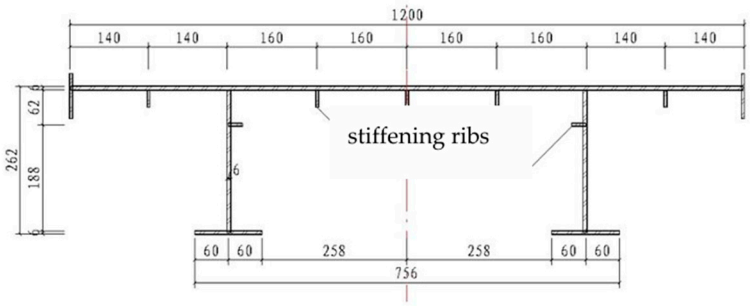
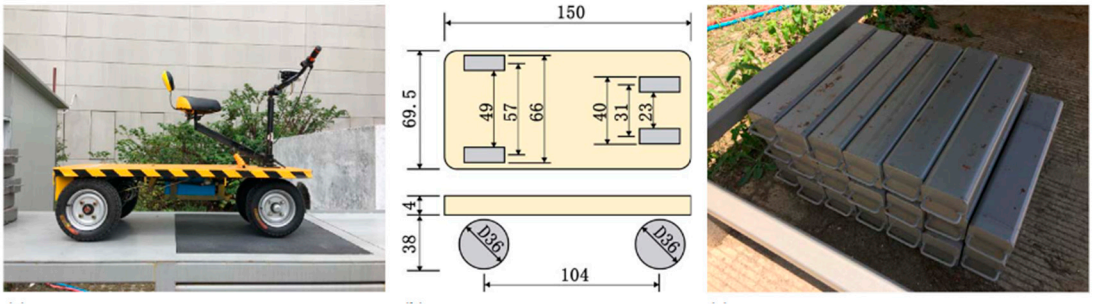


Figure 12. Cross-section of the steel bridge (unit: mm).

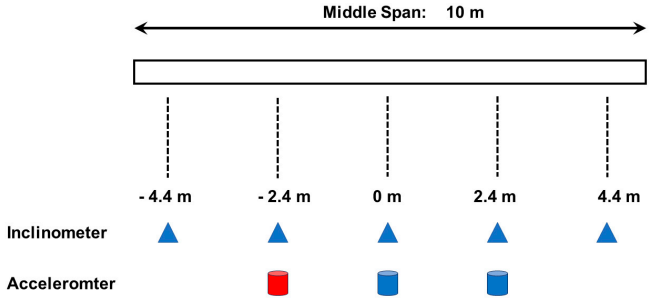
An electric vehicle, as displayed in Figure 13a, was employed to apply a moving load to the bridge. The vehicle's geometric parameters are detailed in Figure 13b. Mass blocks, shown in Figure 13c, were affixed to the vehicle's platform. The front axle weighs 168.5 kg, and the rear axle weighs 218.5 kg. The vehicle traverses the bridge at a constant speed of 0.51 m/s. Throughout multiple trips, the vehicle's path is varied, following random patterns: along the right-side lane, the middle lane, and the left-side lane, as well as navigating an "S" route.

For the monitoring system, five mid-span cross-sections of the bridge have been instrumented with inclinometers at the following coordinates: 0 m (the bridge's geometric center), ±2.4 m, and ±4.4 m. The arrangement of these sensors is depicted in Figure 14a, indicated by triangular and cylindrical markers. The sensors operate at a sampling frequency of 20 Hz. In contrast, accelerometers, which are installed at three cross-sections—specifically at 0 m and ±2.4 m—are also shown in Figure 14a and function at a higher sampling frequency of 200 Hz. Figure 14b provides a detailed view of the sensor placements on each cross-section. The inclinometers, denoted by triangles in Figure 14b, are mounted on the inner side of the bridge's floor plate, while the accelerometers, represented by cylinders, are attached to the outer side. At each cross-section, a single inclinometer is positioned on the left side, with accelerometers installed in symmetrical pairs for balanced data acquisition.

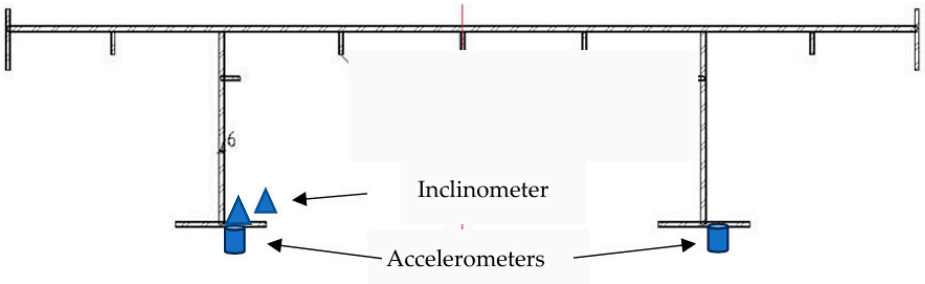


(**a**) (**b**) (**c**)

Figure 13. Electric vehicle (**a**), geometric parameter (unit: cm) (**b**), and mass blocks (**c**).



(**a**) Layout of sensors in the longitudinal direction (the sensors marked with blue denote the known information, while the accelerometer marked with red denotes the acceleration to be estimated).



(**b**) Layout of sensors on the cross-section

Figure 14. Layout of sensors on the middle span (triangle: inclinometer; cylinder: accelerometer).

4.2. Experimental Result Analysis

In this study, the inclination data collected from the five cross-sections were utilized to compute the surrogate inter-section transfer matrix for bridge acceleration. Further, the acceleration data obtained at +2.4 m and 0 m were employed to estimate the acceleration at −2.4 m. It is important to note that, aside from the geometric coordinates of these cross-sections (namely, ±2.4 m, 0 m, and ±4.4 m, along with the support locations), knowledge of other bridge parameters—such as stiffness, density, moment of inertia, and the distance between sensors and the beam's central axis—is not required for the proposed algorithm. Furthermore, information regarding the mass and speed of the electric vehicle is also unnecessary. The method does not rely on finite element models or any prior knowledge as a prerequisite.

A representative set of measured inclination time histories is depicted in Figure 15, revealing that the inclination responses are predominantly influenced by low-frequency components. In this study, we further assumed that the first two vibration modes are the primary contributors to the structural responses. Consequently, the FastICA algorithm was employed to isolate two signal sources and the corresponding mixing matrix derived from the data of the five inclinometers. The separated signal sources are visualized in Figure 16. Given the proximity of the frequencies of these

two sources (by counting the distance between peaks), it is inferred that both sources are composed of the first two principal frequency components, aligning with Equation (18a,b).

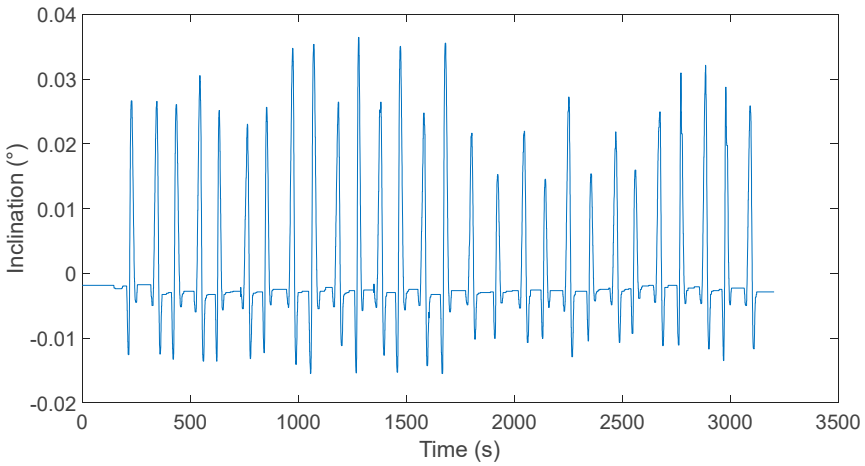
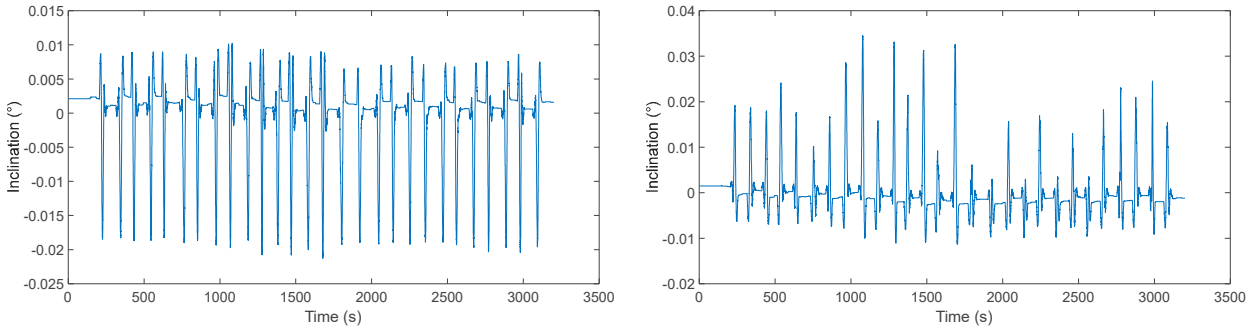


Figure 15. Typical time history of bridge inclination.

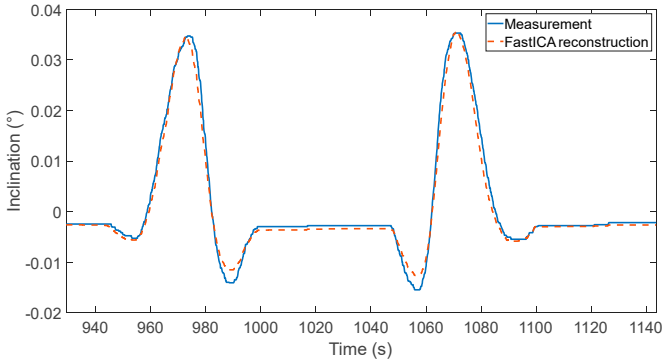


(**a**) Signal source 1 (**b**) Signal source 2

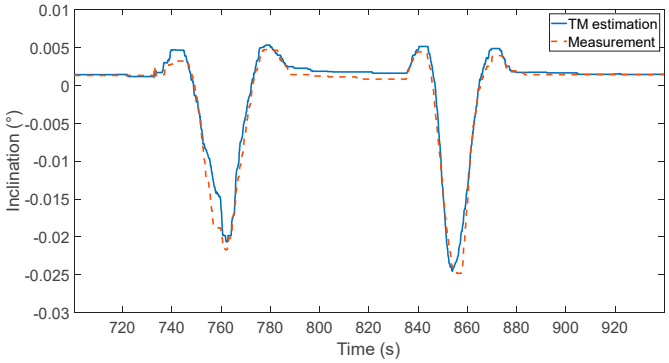
Figure 16. Two separate signal sources with five inclinometers with FastICA.

The accuracy of the signal separation was also scrutinized. Figure 17a presents a comparison between the measured responses at +2.4 m and the reconstructed responses. The reconstructed responses are obtained by multiplying the mixing matrix with the separated signal sources. The comparison demonstrates a close match between the two sets of responses, affirming that the FastICA algorithm effectively isolates accurate signal sources and their corresponding mixing matrices.

Another test of the accuracy of the signal separation was also applied focusing on the mixing matrix. Utilizing the mixing matrix, a surrogate transfer matrix for inclination from the +2.4 m and 0 m cross-sections to the −2.4 m cross-section can be established. In accordance with Equation (12a), elements of the mixing matrix are directly utilized to construct this surrogate matrix. Subsequently, the measured inclination time histories at the +2.4 m and 0 m cross-sections are compiled into an input matrix, which is then multiplied by the surrogate inclination transfer matrix. The result of this operation effectively estimates the inclination time history at the −2.4 m cross-section. Figure 17b, labeled as "TM (transfer matrix) estimation," displays the comparison between the estimated and actual inclination measurements at −2.4 m. The close correspondence between the estimated and measured results substantiates Equation (20b), confirming that the surrogate inter-section inclination transfer matrix is equivalent to the actual one, which validates the second key principle. Additionally, the favorable comparison outcome also suggests that the measured inclination data contains a minimal level of noise.



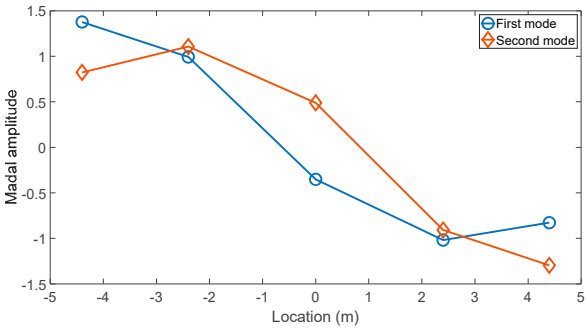
(**a**) Comparison between measurement and reconstruction with FastICA.



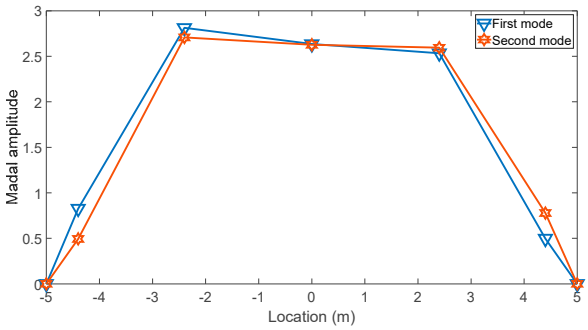
(**b**) Comparison between measurement and reconstruction with inclination transfer matrix.

Figure 17. Comparison between measured inclination, reconstructed inclination from product between mixing matrix and signal sources (shown in (**a**)), and estimated inclination from transfer matrix ("TM estimation in (**b**)").

Having established the accuracy of the FastICA algorithm, the surrogate inclination mode shapes are presented in Figure 18a. Subsequently, the surrogate displacement mode shape function was derived through discrete integration and displayed in Figure 18b. The first and surrogate mode shapes, for both inclination and displacement, appear to be analogous. This similarity is likely due to the fact that the separated signal sources are linear combinations of the actual independent signals with distinct frequencies, as described by Equation (18a,b). Moreover, the corresponding surrogate inter-section transfer matrix of bridge displacement can be obtained.



(**a**) Inclination



(**b**) Displacement

Figure 18. Surrogate inclination mode shape (**a**) and displacement mode shapes (**b**).

Finally, the target acceleration time history at −2.4 m was estimated through acceleration measured at +2.4 m and 0 m with the surrogate transfer matrix of bridge displacement. It is important to note that, given that the high-frequency components predominantly contribute to the acceleration time history, a high-pass filter with a 1 Hz threshold was applied to isolate these high-frequency accelerations. Additionally, the discrete modal amplitudes, as depicted in Figure 18b, on the displace-

ment mode shape curve at ±2.4 m and 0m were utilized to construct the inter-section transfer matrix from +2.4 m and 0m to −2.4 m. Subsequently, Figures 19 and 20 present the comparisons between the estimated and measured accelerations in the time and frequency domains, respectively.

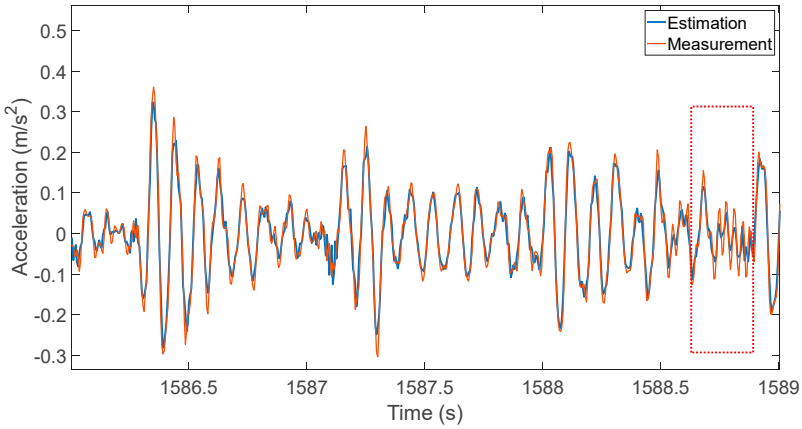
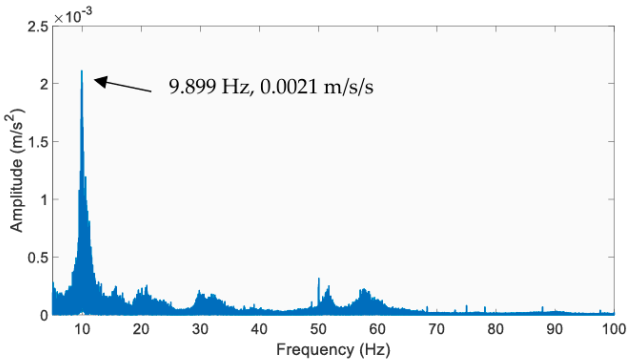
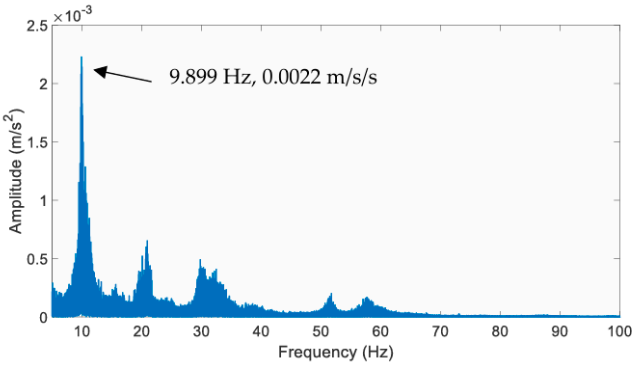


Figure 19. Comparison between the estimated and measured acceleration at −2.4 m. (The blue curve denotes the estimated result while the red curve denotes the measured acceleration. The rectangle marks the high-frequency low-amplitude part.)



(**a**) Frequency spectrum of the estimated acceleration.



(**b**) Frequency spectrum of the measured acceleration.

Figure 20. Comparison of frequency spectrum between estimated and measured acceleration at −2.4 m.

As illustrated in Figure 19, it is evident that in the time domain, the estimated and measured accelerations align closely for the majority of the observed time periods. Despite potential variations in loading conditions across multiple trips, the inter-section transfer matrix is expected to remain consistent for the linear bridge structure. The validation also indicates that the time-varying loading conditions, implicitly contained within the acceleration time histories recorded at +2.4 m and 0 m,

have been effectively translated via the inter-section transfer matrix to accurately estimate the target acceleration at −2.4 m.

However, when the acceleration is characterized by high-frequency, low-amplitude components—highlighted within a red rectangle in Figure 18—the estimated results do not align as closely with the measured accelerations. By checking the frequency spectra of both the estimated and measured accelerations (presented in Figure 20a,b, respectively), for the first vertical mode, the estimated acceleration amplitude is 0.0021 m/s^2, while the measured amplitude is 0.0022 m/s^2, indicating a good match. This concurrence suggests that the vibrations around 10 Hz in the acceleration time history are well correlated between the estimated and measured data. Furthermore, a significant difference is observed in the amplitudes corresponding to higher modes, which is supposed to result in the divergence of high-frequency, low-amplitude vibrations marked in Figure 19. This divergence is likely due to two detailed factors: firstly, the close proximity of the frequencies for the second (20.015 Hz) and third (21.979 Hz) modes of the target bridge, which may present challenges for the algorithm to accurately capture the vibration modes and estimate the inter-section transfer matrix; and secondly, the presence of potential high-frequency noise. In this study, a high-pass filter was solely employed to isolate high-frequency vibration signals, and high-frequency noise was not specifically filtered out, which might result in the observed discrepancies.

Additionally, the limitation of the proposed method is obvious with only the first two modes being considered. In this study, the input acceleration is only measured from two cross-sections; therefore, only the first two modes of vibrations can be considered. Correspondingly, the number of signal resources is designed as two. Furthermore, high-frequency components cannot be estimated accurately. Although Figure 20 illustrates the estimation results around 20 Hz, 30 Hz, 50 Hz, and 60 Hz, this does not necessarily imply that their amplitudes are accurate. This is because the final step (Step 4) of the proposed algorithm involves a linear combination that can indeed preserve these high-frequency components. However, since this study only considers the first two modes, the amplitudes of the higher modes might be incorrectly calculated using the inter-section transfer matrix intended for lower modes. To achieve a more precise estimation of vibrations involving higher modes, it is suggested that additional inputs from more cross-sections would be required. Correspondingly, the signal receiver number, the mixing matrix, and the other parameters involved in the proposed method should also change.

5. Conclusions

In this study, a novel data fusion technology was proposed. By synthesizing the measured low-frequency inclination data across multiple cross-sections with the high-frequency acceleration data from select locations, high-frequency accelerations at the other sections can be estimated. A theoretical analysis was performed to highlight the parallels between the dynamics of a VBI system and the BSS problem. Subsequently, the FastICA algorithm and discrete integration were employed to formulate surrogate inclination mode shapes, surrogate displacement mode shapes, and the surrogate inter-section transfer matrix for acceleration across different bridge cross-sections. Throughout the theoretical derivation, three key principles emerged, elucidating the rationale and methodology behind the application of our proposed data fusion approach. A comprehensive series of finite element simulations and laboratory experiments were executed to substantiate the efficacy of the proposed method. The principal findings are summarized as follows:

(1) High-frequency accelerations and low-frequency displacements share identical mode shape functions and inter-section transfer matrices.
(2) The surrogate inter-section transfer matrix for various parameters, such as displacement and acceleration, derived from the FastICA algorithm, is equivalent to the actual physical inter-section transfer matrix obtained from true physical mode shape functions.
(3) The surrogate displacement mode shape function can be deduced from the discrete integration of the surrogate inclination mode shape function, while the resulting surrogate inter-section transfer matrix for displacement also matches the real physical inter-section transfer matrix for displacement.
(4) A series of finite element models have been constructed to validate the proposed method. The comparison between the estimation results and the simulation data demonstrates that the method in question can yield accurate outcomes for the first two vibration modes, given the current assumption that only these two modes are taken into account. However, the inconsistencies observed in the higher-mode vibrations suggest that the existing set-up is not suitable for estimating the higher-mode components. To attain more precise results, it is recommended that the method be adapted by adjusting the number of signal sources and by

expanding the measurement of acceleration to include more cross-sections, corresponding to the vibration modes under consideration.

(5) A laboratory experiment was also carried out to substantiate the efficacy of the proposed method. Consistent with the validation findings from the numerical analysis, the method exhibits satisfactory performance in estimating lower modes of vibrations. However, it falls short of accurately capturing the higher modes of vibrations under the current configuration, which is limited to considering only the first two modes and relies on acceleration data from just two cross-sections as input. To enhance the precision of higher modes' component estimation, it is imperative to augment the method by incorporating acceleration measurements from an expanded array of cross-sections, along with the requisite adjustments to the associated parameters of the method.

The proposed method is anticipated to eliminate the necessity of creating finite element models. Additionally, it would obviate the need for partial measurement of the target acceleration, significantly reducing the expenditure on sensors and health monitoring equipment. Consequently, with a constrained budget, the proposed approach promises to deliver more extensive monitoring insights.

It is crucial to highlight that within the scope of this study, the influence of measurement noise on the input data for the FastICA algorithm has been found to be negligible. This minimal noise level is essential for ensuring the efficacy of the separated signal sources and the mixing matrix in subsequent applications. However, in real-world engineering contexts, noise conditions may not always be as ideal. Therefore, the implementation of sophisticated noise reduction techniques during data pre-processing could be critical before employing the FastICA algorithm. Moreover, noise levels can also influence the strategic placement of sensors. In this study, sensors were positioned at closely spaced cross-sections. Yet, for long-span bridges where cross-sections are more widely distributed, noise could potentially obscure signals of small amplitude that are indicative of structural responses. Such interference might undermine the effectiveness of the proposed method. Consequently, the sensor layout, particularly the maximum distance between cross-sections, must be meticulously planned to preserve the method's effectiveness. Additionally, the vibration modes to be considered may vary among different bridge types, with long-span bridges requiring a tailored approach. The number of cross-sections from which bridge responses are measured as inputs should be determined before applying the proposed method, which is also related to the layout of sensors. Addressing these challenges will be a priority in forthcoming research.

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